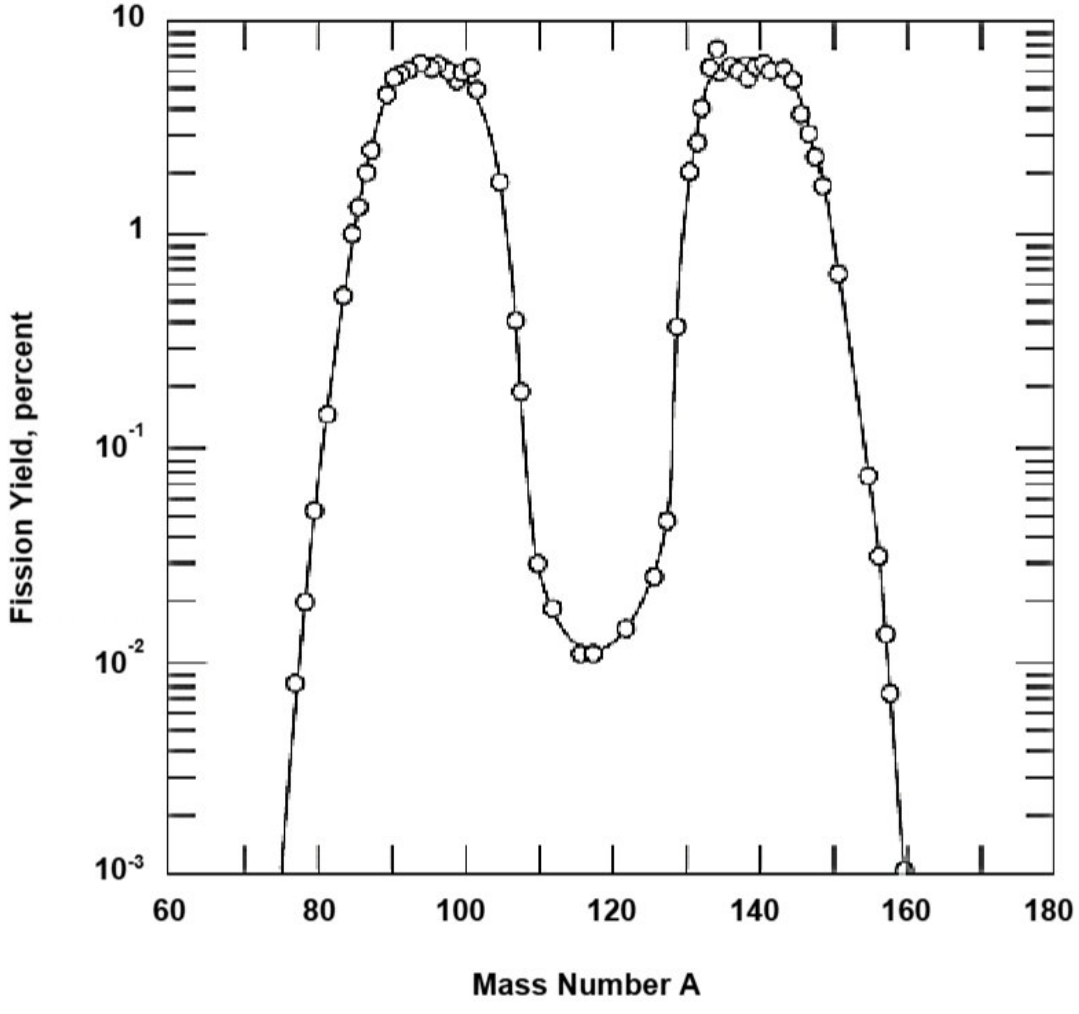
7 When a very strong earthquake of magnitude 9.0 hit Japan on 11 March 2011, the accompanying tsunami caused the devastation of the Fukushima Dai-ichi Nuclear Power Plant (Fk-1). A large amount of radioactive material leaked out of Fk-1 as a result, mainly through cracks made by the quake on some reactors, explosions, intentional vents to relieve pressure, and leakage of cooling water that was contaminated with the melted fuel rod debris.

Fk-1 was capable of generating up to 4.7 gigawatts of power through neutron-induced fissions of Uranium-235 (^{235}U) nuclei. Fission resulting from neutron absorption is called *induced* fission. Some nuclides can also undergo *spontaneous* fission, but that is quite rare. Nuclear fission is a decay process in which an unstable nucleus splits into two fragments of comparable mass, although nearly equal mass is unlikely. Over 100 different nuclides, representing more than 20 different elements, have been found among the fission products. Figure 7.1 shows the distribution of mass numbers for fission fragments from the fission of ^{235}U. Fission yield refers to the percentage of a nuclide of a fission fragment produced per fission. Two or three free neutrons usually appear along with the fission fragments.



Source: https://www.quora.com/Why-is-promethium-produced-from-uranium-235

Figure 7.1

The induced fission of ^{235}U may be represented by a nuclear equation of the form

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}\mathrm{n} \rightarrow {}^{a}_{b}\mathrm{P} + {}^{c}_{d}\mathrm{Q} + (2 \text{ or } 3)\,{}^{1}_{0}\mathrm{n} + \text{energy}$$

where P represents the lower mass number fission fragment, and Q represents the higher mass number fission fragment.

One day after the earthquake, elevated levels of Iodine-131, Cesium-134 and Cesium-137, which are common fission fragments of Uranium-235, were detected near the nuclear plant site. Figure 7.2 shows some general information of the three fission fragments.

Isotope	Symbol	Half-life
Iodine-131	$^{131}_{53}\mathrm{I}$	8.0197 days
Cesium-134	$^{134}_{55}\mathrm{Cs}$	2.065 years
Cesium-137	$^{137}_{55}\mathrm{Cs}$	30.17 years

Figure 7.2

(a) Explain the difference between *induced* and *spontaneous* fission.

.....

.....

..... [1]

(b) Suggest why the *percentage fission yield* in Figure 7.1 is shown on a logarithmic scale.

.....

.....

..... [1]

(c) Using Figure 7.1, state the two most common ranges of mass numbers of the fission fragments (P and Q) of ^{235}U.

most common range of mass numbers of P: ..... [1]

most common range of mass numbers of Q: ..... [1]

(d) **(i)** Uranium-235 undergoes nuclear fission to produce Rubidium-97 and Cesium-137.

Write down the nuclear equation to represent this reaction. Use the chemical symbol Rb to represent Rubidium.

..... [2]

(ii) With reference to Figure 7.1, state the percentage fission yield of Cesium-137 and Rubidium-97.

percentage yield of Cesium-137= .....%

percentage yield of Rubidium-97 = ...% [1]

(iii) Suggest why both the answers in **(d)(ii)** are similar.

.....

.....

..... [1]

(iv) Determine the **two** possible mass numbers of the other fission fragment produced if one of the fragments has a mass number of 81 in the nuclear fission of Uranium-235.

mass number = ... or ..... [2]

(e) Determine which isotope in Figure 7.2 will be about 47% as radioactive after 33 years. Show your reasoning.

isotope: ..... [2]

(f) The release of Iodine-131, Cesium-134 and Cesium-137 into the air, ground and seawater were monitored closely by the Japanese authority because they readily decay by emitting beta-particles and gamma-rays. For example, Cesium-137 decays into Barium-137 (Ba) through beta-emission.

95% of Cesium-137 decays to a metastable excited nuclear energy state (Barium-137m) and is responsible for all emission of gamma-rays, while the other 5% directly decays to the ground state (Barium-137) of the nucleus. Figure 7.4 illustrates the decay scheme of Cesium-137.

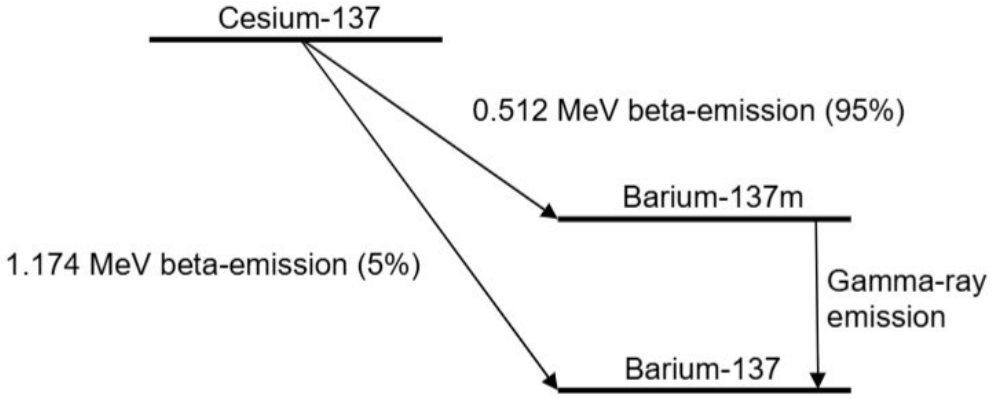
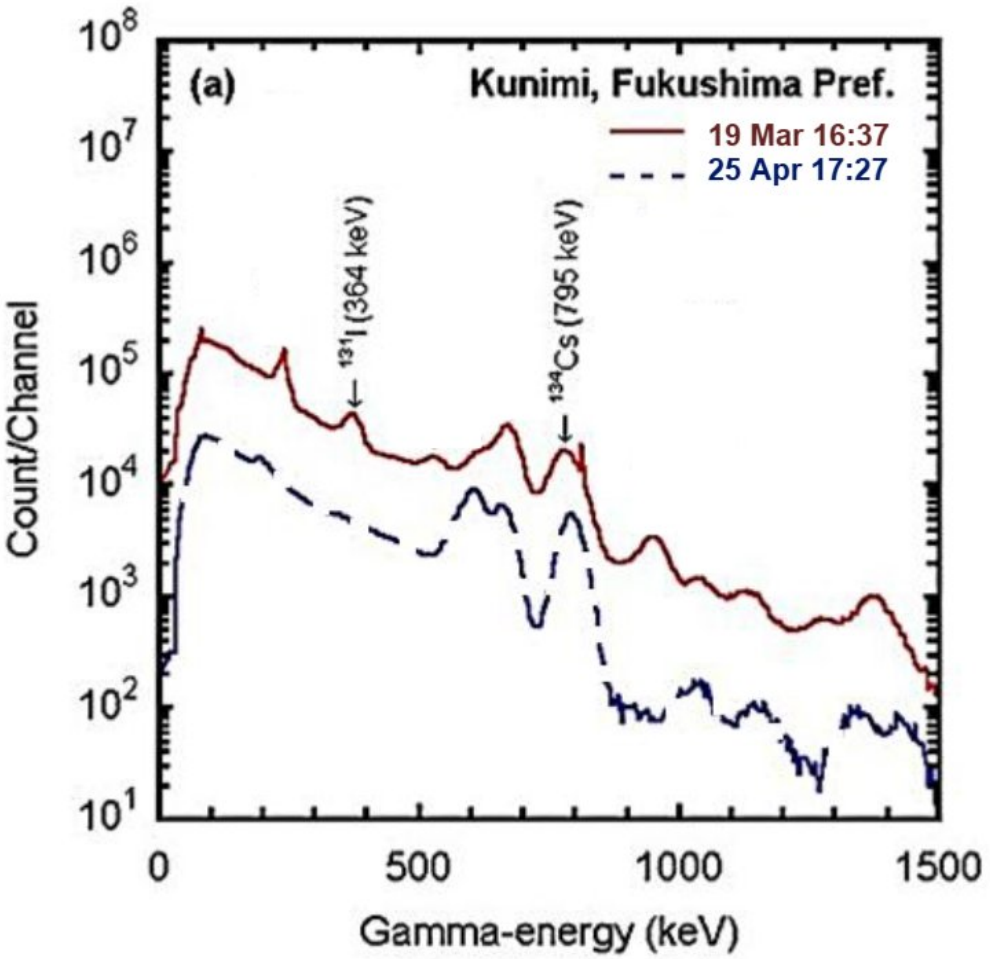


Figure 7.4

Determine the photon energy of the gamma-rays emitted during Cesium-137 decay.

energy = ...MeV [2]

(g) Gamma spectroscopy is a technique used by scientists to identify specific radioactive products from the many possible products that can be produced from the nuclear fission of Uranium-235. Figure 7.5 shows the gamma spectra measured at Kunimi, Fukushima Prefecture on both 19 March 2011 and 25 April 2011. Photon peaks generated by Tellurium-132, Iodine-131, -132, and Cesium-134, -136 and -137 were observed in the spectra given in Figure 7.5. The photon peaks corresponding to Iodine-131 and Cesium-134 are indicated in Figure 7.5.



Source: https://www.nature.com/articles/srep00087 (modified)

Figure 7.5

(i) Using your answer in **(f)**, draw an arrow on Figure 7.5 to show the position of Cesium-137 in the gamma spectra. Label this arrow Cs-137. [1]

(ii) Suggest why no photon peak of Iodine-131 was observed on 25 April 2011.

.....

.....

..... [1]

(h) Compare the three types of radioactive decay: alpha-decay, beta-decay and gamma-decay. Explain which of the three poses the greatest health risk if the source is

(i) outside the body,

.....

.....

.....

.....

..... [2]

(ii) ingested and entered the body.

.....

.....

.....

.....

..... [2]

[Total: 20]

END OF PAPER 2